Diagnostic cancellation and benchmark mismatch in SOD1 folding free energy calculations

Abstract

Alchemical free-energy calculations are increasingly used to estimate how missense mutations change protein folding stability, and their reliability is usually judged with aggregate convergence diagnostics and agreement with experimental controls. We report a validation campaign with internally prespecified gate thresholds for the apo, disulfide-reduced (apo-2SH) monomer of human Cu/Zn superoxide dismutase 1 (SOD1), computed with GROMACS, pmx hybrid topologies and equilibrium λ-window sampling analyzed by MBAR. The gate required Pearson $r \geq 0.70$, RMSE ≤ 1.5 kcal/mol and median net forward/reverse hysteresis ≤ 0.75 kcal/mol against eight charge-conserving experimental controls. It failed: seven usable variants gave $r = 0.326$ and RMSE = 2.123 kcal/mol, and F64A was excluded by a pre-specified per-variant hysteresis cap. Prospective predictions were therefore withheld. Post-verdict audits identified two limitations of the validation procedure itself. First, the historical diagnostic is the absolute value of a sum of signed adjacent-window discrepancies and can conceal large local disagreement: in one F64A folded-leg ladder, net hysteresis was 0.023 kcal/mol while the summed absolute discrepancy was 1.678 kcal/mol (98.6% cancellation). Separately specified continuations of that ladder showed that the small early aggregate did not certify duration stability: the estimate moved by −2.08 kcal/mol between disjoint 3-ns blocks, and repeat movements of opposite sign persisted through 15 ns. Across the folded-leg archive, neither net nor absolute local diagnostics showed a resolvable association with disagreement between independently solvated repeats. Second, the experimental references were not state-matched: all were measured with the Cys57–Cys146 disulfide intact, on cysteine-substituted constructs, and two were whole-dimer calorimetric transitions at 49.4 °C. We report these as limitations of one validation design, not as a validated diagnostic, a structural mechanism, or an accuracy estimate for apo-2SH SOD1.

1. Introduction

Mutations in *SOD1* cause a subset of familial amyotrophic lateral sclerosis, and many disease-associated variants destabilize the immature, metal-free protein [1, 2, 3]. Alchemical free-energy (FE) calculations offer a physics-based estimate of the folding free-energy change of a point mutation, ΔΔG, through a thermodynamic cycle that transforms the residue in a folded protein and in an unfolded-state reference [4, 5]. The approach has been benchmarked extensively on other proteins [4, 5], and the GROMACS/pmx toolchain used here has been applied to SOD1 variants along the maturation pathway, including the apo-2SH state, with non-equilibrium transitions [6]. The present study examines the validation procedure for an equilibrium-window campaign.

Before applying such a calculation to uncharacterized variants, a common safeguard is a validation gate: reproduce experimental controls to a pre-declared accuracy, and require convergence diagnostics to pass. Best-practice guidance emphasizes that diagnostics such as overlap and forward/reverse agreement are necessary rather than sufficient indicators of adequate sampling, and recommends time-resolved checks and independent repeats [7, 8]. Two practical questions are less often reported for a concrete campaign: what a specific aggregate diagnostic can conceal, and whether the experimental controls describe the same thermodynamic state as the simulation.

Here we report a campaign with gate thresholds recorded in the project repository before its first gate evaluation, failed that gate, and then audited both the diagnostic and the benchmark. We preserve the original verdict, exclusion and thresholds, and we label every subsequent analysis as post-verdict and exploratory unless it was separately specified. Our contributions are bounded: (i) a quantified example of signed-discrepancy cancellation in the historical net hysteresis metric, with a grouped analysis showing that neither that metric nor simple absolute alternatives resolved folded-leg repeat disagreement in this archive; (ii) exact-path continuations with separately specified plans showing that a small early aggregate did not certify later duration stability on one selected path; (iii) a chain-of-custody audit showing that the experimental controls differed from the simulated state in redox state, construct, oligomeric transition and temperature; and (iv) two bounded sensitivity results (A4V sampling extension and a G93A disulfide contrast).

2. Methods

2.1 Simulated state and variant panel

The simulated species was the apo, disulfide-reduced SOD1 monomer (153 residues, mature numbering; the precursor N-terminal Met is excluded). This state was chosen before the campaign as the aggregation-prone, metal-free species and to avoid metalloprotein parameterization; it was not chosen to match the experimental controls (Section 3.4). The validation subset comprised eight charge-conserving variants with tabulated experimental ΔΔG: F64A, I18V, I113T, A4V, G93A, G93S, I149A and G93V. Charge-changing variants were excluded because a net-charge change under PME introduces finite-size artifacts that do not cancel between legs with different box sizes, and the engine had no counterion co-alchemy. Experimental values were taken from a published compilation [9] and traced to primary tables (Section 3.4).

2.2 System preparation

The folded leg started from chain A of PDB 3ECU. Disordered internal loops were rebuilt with PDBFixer, heteroatoms including metals and crystallographic waters were removed, and the Cys57–Cys146 disulfide recorded in the deposited structure was broken before protonation at pH 7.0 so that both cysteines carried thiol hydrogens. The wild-type structure was processed with *pdb2gmx*; hybrid residues were generated with *pmx mutate* and *pmx gentop* [10] using the *amber99sb-star-ildn-mut* force field [11, 12] and TIP3P water [13]. *pdb2gmx* disulfide prompts were answered "no" for every cysteine, and a topology guard failed the build if any SG–SG bond was present. The unfolded-state reference was a capped tripeptide (ACE–X–Y–Z–NME) carrying the native flanking residues, with the mutated residue in the central position. Each system was placed in a cubic box with 1.2 nm padding, solvated, neutralized and brought to 0.15 M NaCl, and energy-minimized.

Each of three replicates per leg used an independently solvated and ion-placed system (*independent_replicate_systems*), with deterministic seeds. Independent solvation does not establish independent coverage of protein conformational basins; it removes the shared-box dependence of earlier replicates.

2.3 Alchemical sampling

Each leg used 20 λ states with a refined high-λ endpoint (λ = 0, 0.0588, 0.1176, 0.1765, 0.2353, 0.2941, 0.3529, 0.4118, 0.4706, 0.5294, 0.5882, 0.6471, 0.7059, 0.7647, 0.8235, 0.8824, 0.92, 0.95, 0.975, 1.0). A single λ vector drove both electrostatic and Lennard-Jones terms with Beutler soft-core (α = 0.3, power 1, σ = 0.25 nm, soft-core Coulomb enabled). Because a minimization without the free-energy block relaxes real atoms into appearing dummy atoms, each window was minimized at its own λ before production. Production used the GROMACS stochastic-dynamics integrator (2 fs, collision rate 1 ps^{-1}) at 298.15 K, C-rescale isotropic pressure coupling at 1 atm, PME electrostatics with 1.0 nm real-space and van der Waals cutoffs, long-range dispersion corrections, and LINCS constraints on bonds to hydrogen. Reduced potentials at all 20 states were written at a stride targeting 3,000 retained records per window (3,001 in the surviving attempt-1 archive because the boundary record was retained). GROMACS 2025.3 (CUDA build) [14] ran one window per NVIDIA L40S GPU.

In gate attempt 1 (protocol identifier *822108e9db71124d*), each window of both legs used 0.5 ns discarded equilibration and 3 ns production (8 variants × 2 legs × 20 states × 3 replicates = 960 windows). A single documented protocol change for attempt 2 (S1 Appendix, Section S1) extended the folded leg of A4V only to 2 ns equilibration and 9 ns production; the unfolded windows were reused unchanged. Every window recorded its engine and a hash of its production parameters, and the analysis refused to combine windows with differing hashes or engines.

2.4 Free-energy estimation and the historical diagnostic

For each leg and replicate, every window's reduced-potential series was trimmed at the later of two automatically detected equilibration times — for the sampled-state energy and for the neighbor energy difference — and thinned at the larger of the two statistical inefficiencies [15]. Leg free energies were estimated by MBAR [16] with pymbar. ΔΔG was the folded-minus-unfolded difference averaged over replicates; its reported uncertainty was the larger of the propagated MBAR error and the replicate standard error of the mean. That uncertainty is conditional on the sampled states; it does not include force-field or reference-state error or unvisited conformational modes.

The historical diagnostic, termed "cycle closure" in the project code, is a forward/reverse hysteresis of one λ ladder, not a closed thermodynamic cycle. For adjacent states k and k+1, let d_k be the difference between the forward exponential-averaging estimate (samples from k) and the reverse estimate (samples from k+1). The ladder hysteresis is

$$H = |\Sigma_k d_k|,$$

and each variant's value was the maximum H over its six leg–replicate ladders. Because signed d_k can cancel, we also report the descriptive summaries $S = \Sigma_k |d_k|$ and $M = \max_k |d_k|$, and the cancellation fraction $1 - H/S$. These depend on the number and placement of λ states; all comparisons here use the same 20-state schedule.

2.5 Internally prespecified gate

Thresholds appear in a repository commit dated 2026-08-07 that precedes the recorded first gate verdict: Pearson $r \geq 0.70$, RMSE ≤ 1.5 kcal/mol, median per-variant hysteresis ≤ 0.75 kcal/mol, at least six usable variants, and a per-variant hysteresis cap of 1.0 kcal/mol for exclusion. Spearman ρ was reported without a threshold. A pre-declared pivot applied if r fell below 0.60: reframe the work as a methods and sampling-limits study rather than retune. Failure of the gate prohibited predictions for uncharacterized variants, and this prohibition was enforced in code. The repository chronology documents internal prespecification, not an independently timestamped public preregistration. Protocol details changed during development; the evidence does not establish that the entire campaign protocol was frozen on that date (S1 Appendix, Section S3).

2.6 Post-verdict audits

All analyses in this section were defined after the gate verdict and are exploratory unless stated otherwise. None revises the verdict.

Archive re-derivation. All 48 retained (variant, leg, replicate) ladders (912 adjacent pairs) were re-solved from the archived reduced-potential files. A4V's attempt-1 folded windows had been overwritten by the attempt-2 run, so its gate point survives only as a run-time record; its retained ladders are the 9-ns folded data.

Grouped diagnostic association. For each folded and unfolded ladder we computed the disagreement with its two independently solvated siblings, $|\Delta G_i - \text{mean}(\Delta G_j, j \neq i)|$, and its association with H, S, M, minimum

adjacent overlap and conditional MBAR error. Because disagreements reuse sibling estimates, rows are dependent; 24 folded ladders are nested in eight variants at six sites. We therefore report Pearson and Spearman coefficients with 95% percentile intervals from 2,000 whole-variant and whole-site bootstrap draws, leave-one-group- out values, and a separate seven-variant attempt-1 cohort excluding A4V. No row-wise p-values are reported.

Retrospective temporal analysis. Each attempt-1 window was split into four non-overlapping quarters. Estimates from cumulative prefixes ending at 25%, 50% and 75% were compared with an estimate from the final quarter alone, recomputing trimming and thinning for every interval.

F64A continuations with separately specified plans. Two separately specified folded-only continuations extended the identical F64A apo-2SH paths (same 20 states × 3 repeats; no new starts, velocities or unfolded legs): v1 from 3 to 9 ns of retained sampling and v2 from 9 to 15 ns. The v2 primary endpoint was the paired movement from the retained 9–12 ns block to the 12–15 ns block under the adaptive trimming policy, reported alongside two fixed policies (untrimmed with thinning; fixed 25% trim with thinning) that reuse the same data. No structural trajectories were retained under the production output policy. A further descriptive audit computed neighbor and fixed λ-pair reduced-energy drift against actual λ from all 60 v2 energy files.

Benchmark-state audit. Each control value was traced from the compilation to its primary table, recording transition, redox state, metal state, construct, temperature, pH and method. A post-verdict sensitivity analysis varied only the normalization and temperature treatment of the two dimer calorimetric controls, keeping every prediction and the exclusion fixed.

G93A disulfide contrast. A separately specified diagnostic repeated the G93A attempt-1 protocol with the Cys57–Cys146 disulfide intact (three independently solvated folded and unfolded systems), verified the redox state of each arm by parsing atom and bond tables, and compared folded ΔG between arms with an unpaired Welch interval.

2.7 Reproducibility

CPU-only workflow targets re-derived estimates and diagnostics from the surviving energy archives. The gate-of-record comparison preserves A4V's attempt-1 estimate from its historical run-time record: its folded raw

windows were overwritten and cannot be re-derived. The source map distinguishes this preserved value from estimates regenerated from raw energies. The F64A temporal figure builder checks its source-report hashes and specified quoted values; supplementary images are checked against pinned hashes before copying. A forensic release bundles the historical archive manifest, configuration, analysis code and figure builders; the later continuation reports and manuscript figure assembly are separate repository artifacts (Data availability).

Software versions are recorded by analysis rather than assigned retrospectively to all runs. The local forensic reproduction environment is pinned in *env/analysis-lock.txt* (Python 3.13.9, pymbar 4.0.3, NumPy 2.3.5 and SciPy 1.16.3). The saved F64A v1 block and v2 continuation reports instead record Python 3.11.15, pymbar 4.2.0, NumPy 2.4.6, and JAX/jaxlib 0.10.2 on SCC. The original gate environment specification required pymbar ≥ 4 without an exact pin; neither the local lock nor the later continuation versions establishes its original runtime version. This is a historical software-provenance limitation. Recorded per-analysis identities are retained with the reports.

3. Results

3.1 The prespecified gate failed

The gate failed on correlation and error (Table 1; Fig 1). Seven usable variants gave Pearson r = 0.326, Spearman ρ = 0.500, RMSE = 2.123 kcal/mol and mean unsigned error 1.785 kcal/mol; median hysteresis (0.22 kcal/mol at evaluation; 0.225 when re-derived from the archived windows) and the point count passed. F64A was excluded because its maximum ladder hysteresis was 1.104 kcal/mol, above the pre-specified 1.0 kcal/mol cap; the exclusion followed the prespecified criterion, not the value of its estimate. Pearson r fell below the 0.60 pivot line, so the work was reframed as reported here and no uncharacterized variant was simulated.

Fig 1. Gate of record. Computed versus experimental ΔΔG for the seven usable controls of attempt 1, with the excluded F64A marked by its pre-specified hysteresis exclusion. Experimental references are not state-matched to the simulation (Section 3.4).

Table 1. Gate of record (attempt 1). Experimental values are the compilation values used by the gate (kcal/mol, positive = destabilizing). FEP values are mean ± reported uncertainty over three replicates.

Variant	Experimental	FEP	Error	Max. ladder H
I18V	0.37	0.83 ± 0.21	+0.46	0.18
I113T	1.25	2.59 ± 0.23	+1.34	0.22
A4V	1.62	3.54 ± 0.39	+1.92	0.66
G93A	2.43	1.26 ± 0.06	−1.17	0.04
G93S	3.70	1.20 ± 0.07	−2.50	0.12
I149A	4.05	4.98 ± 0.16	+0.93	0.77
G93V	7.00	2.83 ± 0.08	−4.17	0.67
F64A	−0.20	6.94 ± 0.34	excluded	1.10

Errors were positive for all four non-glycine variants and negative for all three G93 substitutions; the experimental span at position 93 was 4.57 kcal/mol, and the computed span 1.63 kcal/mol. We record this pattern without assigning a cause: Section 3.4 and S1 Appendix show that the comparison combines calculation error with benchmark heterogeneity, and the present data cannot separate them. The gate justified withholding predictions. It does not measure accuracy against matched apo-2SH experiments or isolate force-field error.

All documented attempt-1 estimates except A4V's gate point re-derive from the archived windows to floating-point rounding; A4V's 3-ns folded windows were overwritten in place by the attempt-2 run, and no later analysis can repair that missing record.

3.2 The net hysteresis metric cancels signed local discrepancies

H can be small while adjacent forward/reverse estimates disagree substantially, because d_k of opposite sign cancel in the sum. In F64A folded replicate r1 at 3 ns, H = 0.022705 kcal/mol, whereas S = 1.678103 and M = 0.222590 kcal/mol, a cancellation fraction of 98.647% (Fig 2). The same ladder had the largest disagreement with its independently solvated siblings in the archive, 1.202554 kcal/mol. Cancellation concerns the reported

diagnostic; it does not show that cancellation caused the MBAR estimate to be wrong, or that either sibling is correct.

Fig 2. Signed local discrepancies and alternative summaries. Top left: signed adjacent forward/reverse discrepancies d_k along the 20-state ladder for F64A folded r1 at 3 ns (H = 0.023, S = 1.678, M = 0.223 kcal/mol; 98.6% cancellation). Other panels: sibling disagreement against H, S and M for all 24 folded ladders (eight variants, six sites; points are dependent). Squares are A4V's retained 9-ns ladders; red marks F64A r1.

Across the archive, the historical metric did not track sibling disagreement on the folded leg: over 24 folded ladders, Pearson r = +0.072 and Spearman ρ = +0.080 (Fig S1). The descriptive alternatives did not resolve an association either (Table 2). Every folded-leg 95% interval, under both variant and site resampling and in both cohorts, included zero. On the unfolded leg, conditional MBAR error was positively associated with disagreement (r = 0.421; variant-bootstrap interval 0.087 to 0.646; site-bootstrap 0.025 to 0.609), and the variant-grouped Pearson interval for H narrowly excluded zero (0.020 to 0.759) while its site-grouped interval did not. With so few groups these intervals are approximate, and the near-zero folded correlations do not establish an absence of diagnostic information in a wider population. We select no threshold from the F64A example.

Table 2. Association between ladder diagnostics and sibling disagreement, folded leg. Point estimates over 24 ladders (eight variants, six sites); 95% percentile intervals from 2,000 site-bootstrap draws.

Diagnostic	Pearson r [95% site interval]	Spearman ρ [95% site interval]
Net hysteresis H	+0.072 [−0.29, +0.16]	+0.080 [−0.31, +0.32]
Sum of absolute discrepancies S	+0.104 [−0.48, +0.34]	+0.162 [−0.49, +0.47]
Maximum local discrepancy M	+0.158 [−0.45, +0.28]	+0.136 [−0.61, +0.42]
Minimum adjacent overlap	−0.120 [−0.25, +0.65]	−0.189 [−0.31, +0.55]
Conditional MBAR error	+0.233 [−0.43, +0.68]	+0.017 [−0.47, +0.65]

Overlap did not order the gate errors either. The two thinnest ladders in the archive (I18V, minimum adjacent overlap 0.0054; I149A, 0.0096) belonged to the two variants with the smallest gate errors, and G93A had the lowest variant hysteresis (0.045) but an error of −1.17 kcal/mol.

3.3 A small early aggregate did not certify duration stability

The retrospective quarter analysis showed that attempt-1 estimates were still moving within the retained 3 ns. For folded ladders, the median absolute difference between the 75% prefix and the final quarter was 0.311 kcal/mol (maximum 1.070). The median of the 21 per-record ratios of absolute movement to combined conditional MBAR uncertainty was 2.43, with each denominator calculated as the square root of the sum of the two squared MBAR errors; 12 of 21 primary-cohort records exceeded a ratio of two. This quadrature scale does not estimate covariance between the blocks. At the 25% prefix the folded median was 0.368 kcal/mol (maximum 1.808, F64A r1). These are descriptive scale comparisons on dependent data, not tests.

The v1 continuation extended all three F64A folded paths from 3 to 9 ns (Fig 3a). Replicate r1 moved from 2.505942 kcal/mol in its first retained 3-ns block to 0.429467 kcal/mol in the disjoint 6–9 ns block, a change of −2.076474 kcal/mol, despite an early net hysteresis of 0.022705 kcal/mol. The other two repeats moved by −0.691 and −0.584 kcal/mol between the same blocks. The cumulative mean changed from 1.704 to 0.771 kcal/mol between 3 and 9 ns, and the repeat range narrowed by 71%. The small early net hysteresis coexisted with the −2.076474 kcal/mol movement between the 0–3 and 6–9 ns blocks (Fig 3b). This is an exact-path counterexample to treating a small aggregate as a duration-stability certificate. It is a selected case, not a prospective validation cohort, and neither block is an established equilibrium value.

Fig 3. F64A folded-leg temporal behavior on exact-path continuations. (a) Folded-leg ΔG estimates in disjoint 3-ns retained blocks for three repeats (adaptive policy; error bars, conditional MBAR uncertainty, which excludes unvisited conformations). v1 covers 0–9 ns and v2 9–15 ns; the 6–9 ns block is identical in both reports. (b) Net hysteresis H for the same blocks. (c) Primary v2 paired movement from 9–12 to 12–15 ns for each repeat under the three saved policies, which reuse the same data; bars are policy means. (d) Adaptive retained suffix length per window against actual λ in the 12–15 ns block. Values are not equilibrium free energies or experimental comparisons.

The v2 continuation extended the same paths to 15 ns. Under the adaptive policy, the primary paired movement from 9–12 to 12–15 ns averaged −0.049510 kcal/mol, but this small mean reflected opposing repeat movements: r0 −0.051, r1 −0.695174 and r2 +0.598142 kcal/mol (Fig 3c). Both fixed policies gave upward

mean movements, +0.579425 (untrimmed) and +0.410739 kcal/mol (fixed 25% trim), from the same raw data. The adaptive repeat range was 2.047618 kcal/mol in the 9–12 ns block and 1.946359 kcal/mol in the 12–15 ns block. Repeat estimates therefore did not agree, and their movement depended on the selection policy, through 15 ns.

Adaptive selection was strongly time-local at the highest λ values (Fig 3d; Fig S4). In the 12–15 ns block, the selected suffix for r2 at λ = 0.95 kept 32 of 3,000 raw records, leaving seven samples after thinning. Sparsity was not confined to the adaptive cutoff: in the same block both fixed policies retained 10 or fewer samples for r1 at that window, with estimated statistical inefficiencies near 330. Short selected suffixes and a quiet aggregate cannot certify equilibrium.

A descriptive audit of all 60 v2 energy files found larger neighbor-energy drift toward high λ. Taking window medians over repeats, blocks and directions, the median absolute half-block drift was 0.0223 kT for windows 0–9 and 0.0843 kT for windows 10–19 (Spearman ρ with window index, 0.848), and the pattern was present in each repeat separately (Figs S2–S3). This localizes an energy observable. It does not establish a structural mechanism, cavity hydration or that low-λ paths are equilibrated; window 9 lies at λ = 0.53, so low-index windows are not wild-type-like controls. The F64A hybrid removes six ring heavy atoms while retaining CB; cavity hydration or repacking around that site remain hypotheses, untested here because no structural trajectories were retained.

3.4 The experimental controls were not state-matched

All eight gate values traced to primary tables (Table 3). Six — A4V, G93A and I113T [1]; F64A, I18V and I149A [2] — are monomer urea-unfolding measurements on a C6A/C111A/ F50E/G51E construct in 10 mM MES, pH 6.3, at 25 °C. Both sources state that the Cys57–Cys146 disulfide was kept intact. The remaining two, G93S and G93V [17], are apo native-dimer differential scanning calorimetry results on a C6A/C111S construct at pH 7.8, analyzed as a cooperative two-unfolded-monomer transition and reported at a common 49.4 °C, also under non-reducing conditions. The simulation used the reduced, native-sequence monomer, prepared at pH 7.0 and simulated at 298.15 K. No control was fully matched.

Table 3. Experimental-state compatibility of the gate controls.

Attribute	Six monomer controls	Two dimer controls	Simulation
Variants	A4V, G93A, I113T; F64A, I18V, I149A	G93S, G93V	All eight
Primary tables	Lindberg 2005, Table 1; Nordlund 2006, Table 1	Stathopulos 2006, Table 2	This work
Transition	Monomer unfolding	Native dimer → 2 unfolded monomers	Monomer mutation cycle
Redox state	Disulfide intact (SS)	Disulfide intact (SS)	Reduced (2SH)
Construct	C6A/C111A/F50E/G51E	C6A/C111S	Native sequence
Temperature	25 °C	49.4 °C	25 °C
pH	6.3	7.8	7.0 at preparation
Method	Urea unfolding	Calorimetry (DSC)	MD/MBAR

The compilation's uniform ±0.3 kcal/mol [9] is a blanket methodological estimate across sources, not a per-point measurement error. Opposite-signed residuals reject a single additive offset between states, but not mutation-dependent responses to the same state or construct change.

Because G93S and G93V carry the largest gate residuals, the aggregate metrics were sensitive to how those two references are treated (Fig 4). Converting both to 25 °C and halving the dimer transition, with every prediction fixed, gave r = 0.754 and RMSE = 1.074 kcal/mol under a constant heat-capacity assumption, or r = 0.767 and RMSE = 1.117 with a temperature-dependent heat capacity. Neither scenario is a gate pass. The scenarios were constructed after the verdict; halving a cooperative dimer transition does not make it an isolated-monomer measurement or match construct and redox state; and naive propagation of the reported absolute uncertainties gives about ±1.2 kcal/mol per halved 25 °C reference, not ±0.3. The analysis shows only that the failed gate measured calculation error together with benchmark heterogeneity.

Fig 4. Reference sensitivity with fixed predictions. Frozen predictions against four reference constructions that differ only in G93S/G93V: gate of record, dimer normalization only, 25 °C conversion only, and both (constant heat capacity). Temperature-dependent heat-capacity scenarios are reported in the text. Post-verdict scenarios, not gate evaluations.

3.5 Supporting sensitivity studies

Two bounded sensitivity studies place limits on the interpretation of the gate errors (S1 Appendix, Sections S1–S2; Figs S5–S6). Extending A4V folded sampling while also increasing equilibration narrowed repeat spread but left the discrepancy from its unmatched reference. A G93A disulfide contrast gave a folded free-energy difference of −0.017 kcal/mol with an unpaired 95% interval of −0.297 to 0.263 kcal/mol. Neither comparison identifies the cause of the gate residuals or establishes adequate sampling of slower conformational changes.

4. Discussion

The campaign did what a prespecified gate is for: it stopped prospective predictions when the calculation did not reproduce its controls. The more transferable findings concern why a gate of this design can be hard to interpret in either direction.

Aggregate diagnostics. The historical metric took an absolute value after summing signed local discrepancies, so it could report near-perfect agreement for a ladder whose local forward/reverse estimates disagreed by a cumulative 1.7 kcal/mol, and whose estimate later moved by 2.1 kcal/mol. Reporting signed local discrepancies, their absolute sum and their maximum alongside any aggregate costs nothing once all-state energies are saved. We do not, however, resolve an association between these alternatives and folded-leg repeat disagreement in this archive, and we do not propose a threshold. The appropriate conclusion is narrower: an aggregate that permits cancellation should not be presented as evidence of convergence on its own, consistent with existing guidance [7, 8].

Duration and selection. Small early hysteresis preceded a 2.08 kcal/mol difference between the 0–3 and 6–9 ns blocks on the selected F64A path. Through 15 ns the paired movement depended on the trimming policy while repeats moved in opposite directions. Adaptive equilibration detection [15] sometimes retained only a few late samples in high-λ windows. A near-zero mean movement is therefore not evidence of stability when the repeats disagree, and policy dependence should be reported rather than resolved by choosing a preferred policy after the data are seen. Independent solvation of repeats is useful but does not establish independent conformational coverage; denser λ placement can improve statistical bridging between states but does not by itself accelerate slow dynamics within unchanged states.

Benchmark state. Every control was measured on a different chemical species than the one simulated (disulfide intact, cysteine-substituted constructs), and two were whole-dimer calorimetric transitions at a different temperature. Uniform compilation uncertainties concealed this. A failed comparison against such a benchmark does not show that apo-2SH FE calculations are inaccurate, and a passing comparison would not have shown that they are accurate. For future validation we recommend recording, for each control, the transition, redox and metal state, construct, temperature, pH, method and source table before selecting the panel; we did not assemble a matched apo-2SH benchmark here.

Practical recommendations. (1) Preserve signed local diagnostics alongside aggregate values. (2) Represent the experimental transition and state explicitly when choosing a validation panel. (3) Keep conditional sampling uncertainty, repeat spread and discrepancy from experiment separate. (4) Archive immutable run identities, topologies and raw energies before any resubmission; the A4V attempt-1 windows were lost by overwriting, and a parameter hash alone does not certify physical identity. (5) Distinguish statistical bridging between λ states from conformational exploration within states, and compare fixed observables when schedules differ. (6) Evaluate any proposed stopping rule on external systems before treating it as general.

Limitations

The archive contains eight variants at six sites with three repeats each, so grouped intervals are wide and approximate, and diagnostic associations cannot be generalized. Disagreement values reuse sibling estimates and are dependent. The temporal counterexample is a single selected path; continuation blocks remain correlated through slow modes; the adaptive and fixed policies reuse the same data. Conditional MBAR uncertainties exclude unvisited conformations. No structural trajectories were retained, so no structural interpretation is offered. One force field, one water model, one starting structure and one soft-core form were used. The unfolded reference is a capped tripeptide whose adequacy as an unfolded-state model was not tested. A4V's attempt-1 folded windows are lost. The G93A contrast covers one variant at 3 ns. The benchmark audit identifies incompatibility; it does not estimate the size of any state- dependent correction for the simulated species.

5. Conclusions

An internally prespecified apo-2SH SOD1 folding-stability gate failed against a heterogeneous experimental compilation and stopped prospective predictions. Its convergence diagnostic could hide substantial signed local disagreement, a small early value did not certify later duration stability on a separately specified continuation, and its experimental controls differed from the simulated species in redox state, construct, oligomeric transition and temperature. These findings describe limitations of the validation procedure. They do not identify the physical cause of the prediction errors, validate an alternative diagnostic, or establish an equilibrium free energy or an accuracy estimate for apo-2SH SOD1.

Data availability

The analysis code, configuration, surviving reduced-potential archives and reports are assembled in a local deposit candidate. A public repository accession has not yet been assigned. A4V attempt-1 folded raw windows were overwritten; only their historical run-time estimates and uncertainty survive. The deposit inventory distinguishes that exception from reproducible raw-energy analyses. Structural trajectories were not retained. The source inventory and reproduction instructions are described in S1 Appendix, Section S4.

References

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Supporting information

S1 Appendix. Sensitivity studies and evidence provenance. A4V sampling comparison, G93A disulfide contrast, gate chronology, reproducibility inventory and Figs S1–S6.

Fig 1 | SOD1 validation study

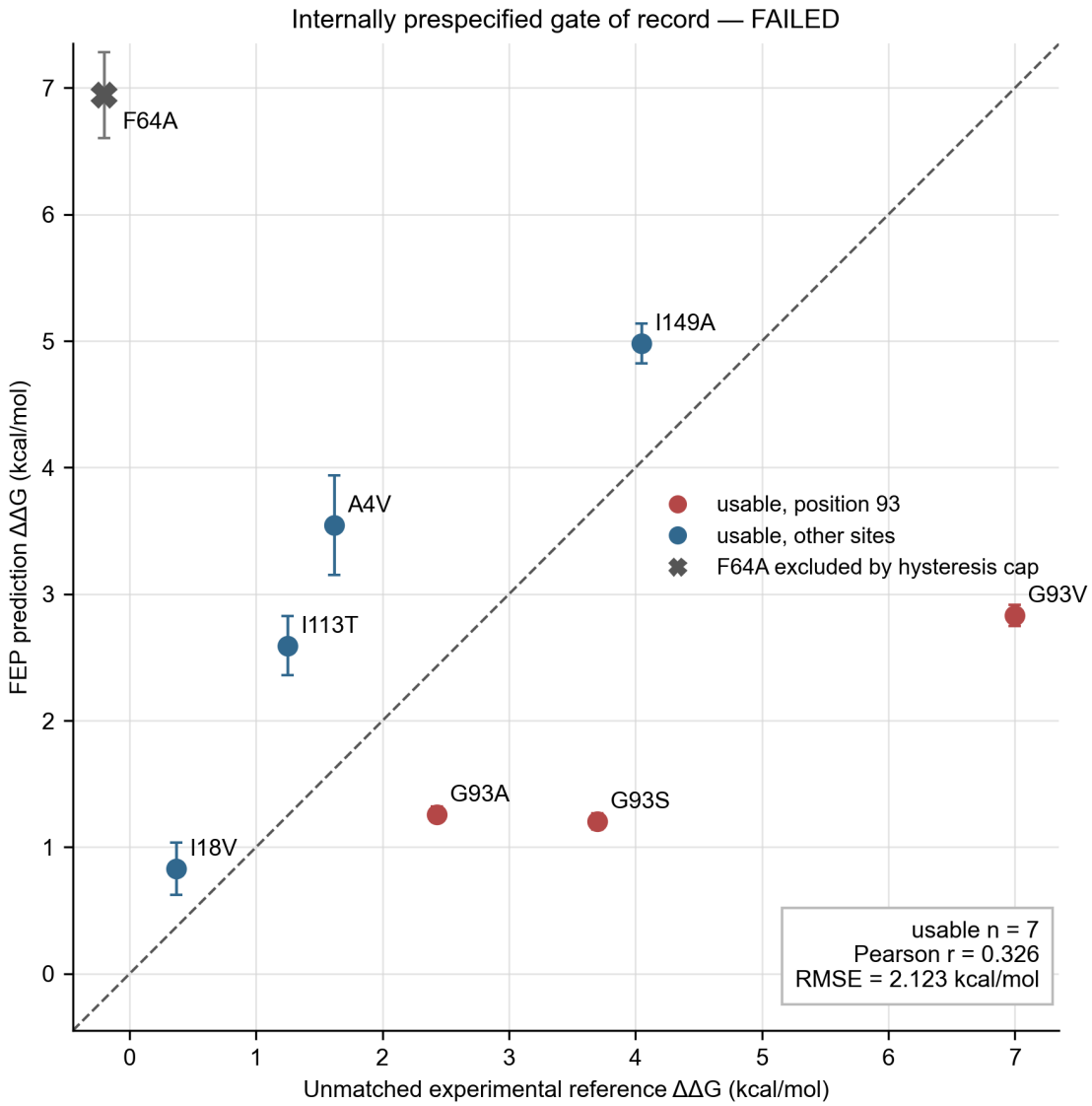


Fig 1. Gate of record. Computed versus experimental ΔΔG for the seven usable controls of attempt 1, with the excluded F64A marked by its pre-specified hysteresis exclusion. Experimental references are not state-matched to the simulation (Section 3.4).

Fig 2 | SOD1 validation study

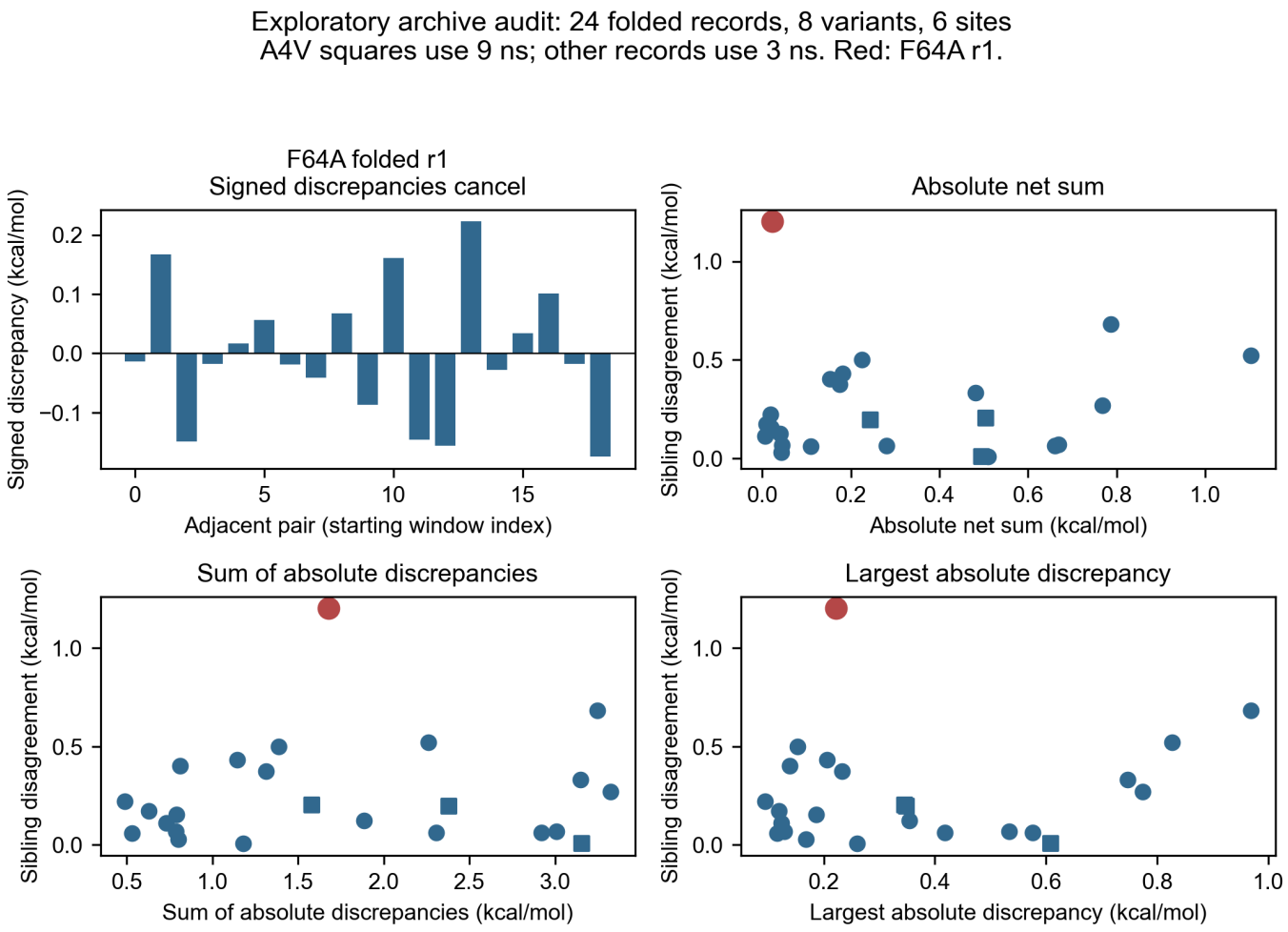


Fig 2. Signed local discrepancies and alternative summaries. Top left: signed adjacent forward/reverse discrepancies d_k along the 20-state ladder for F64A folded r1 at 3 ns (H = 0.023, S = 1.678, M = 0.223 kcal/mol; 98.6% cancellation). Other panels: sibling disagreement against H, S and M for all 24 folded ladders (eight variants, six sites; points are dependent). Squares are A4V's retained 9-ns ladders; red marks F64A r1.

Fig 3 | SOD1 validation study

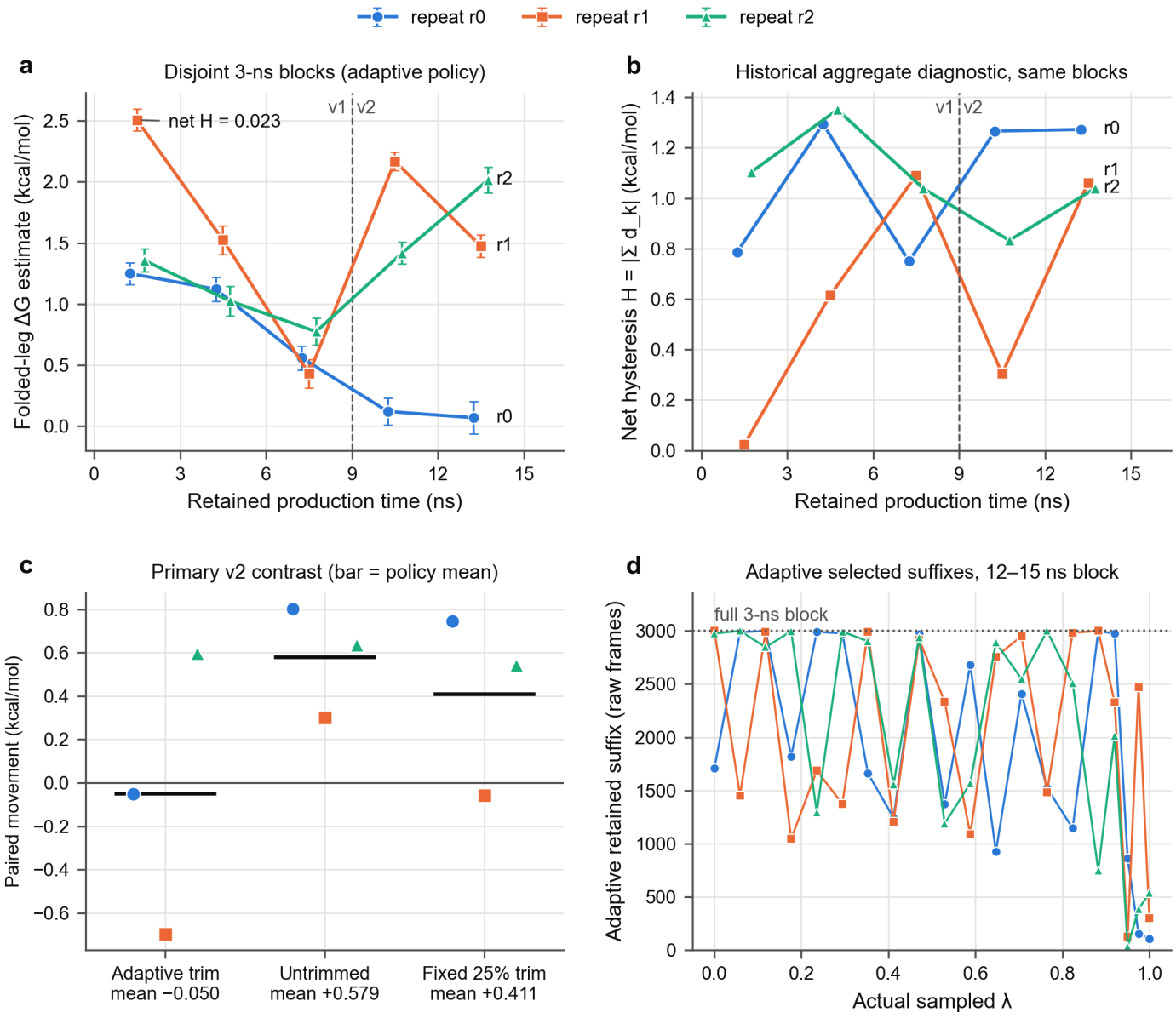


Fig 3. F64A folded-leg temporal behavior on exact-path continuations. (a) Folded-leg ΔG estimates in disjoint 3-ns retained blocks for three repeats (adaptive policy; error bars, conditional MBAR uncertainty, which excludes unvisited conformations). v1 covers 0–9 ns and v2 9–15 ns; the 6–9 ns block is identical in both reports. (b) Net hysteresis H for the same blocks. (c) Primary v2 paired movement from 9–12 to 12–15 ns for each repeat under the three saved policies, which reuse the same data; bars are policy means. (d) Adaptive retained suffix length per window against actual λ in the 12–15 ns block. Values are not equilibrium free energies or experimental comparisons.

Fig 4 | SOD1 validation study

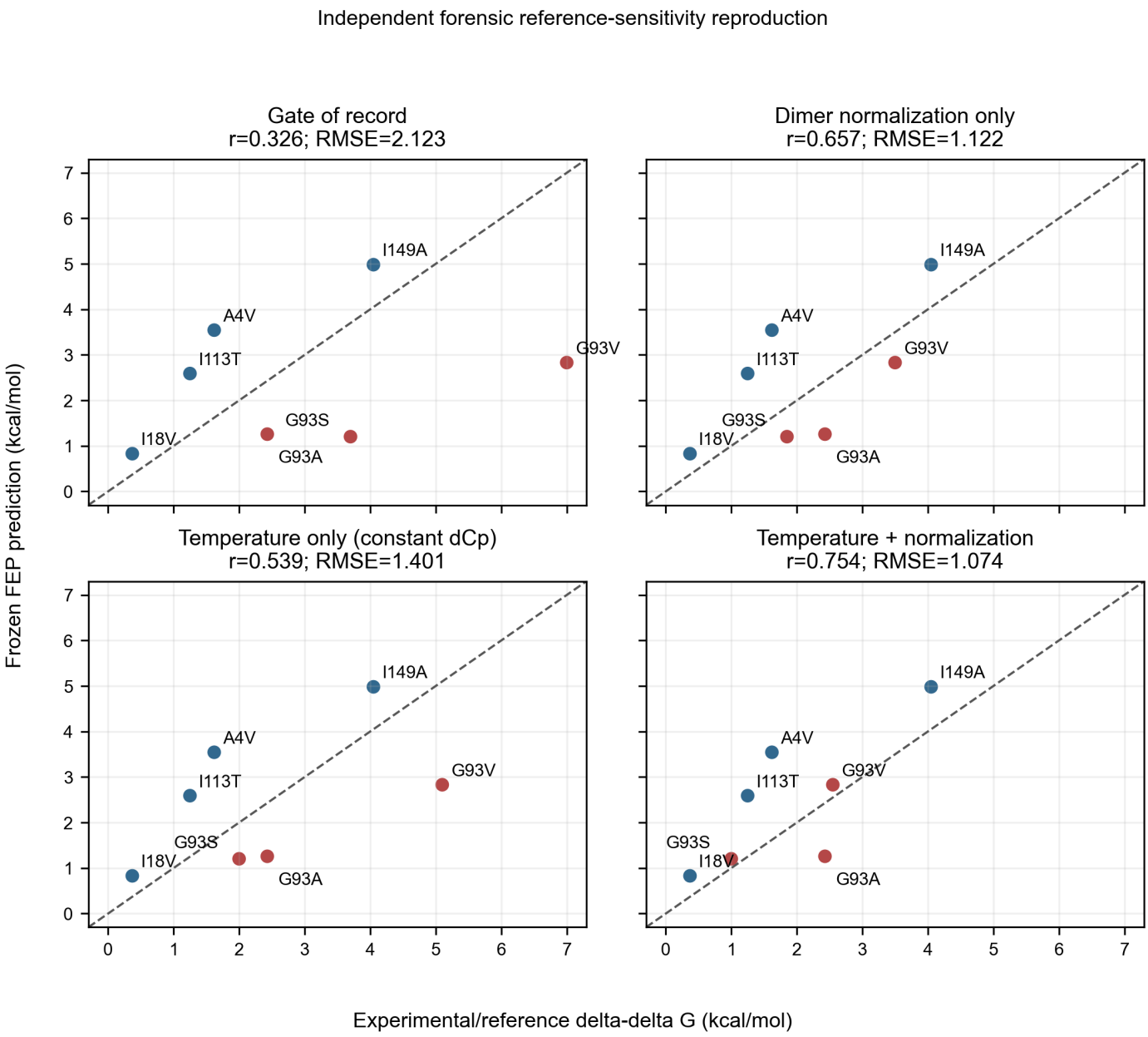


Fig 4. Reference sensitivity with fixed predictions. Frozen predictions against four reference constructions that differ only in G93S/G93V: gate of record, dimer normalization only, 25 °C conversion only, and both (constant heat capacity). Temperature-dependent heat-capacity scenarios are reported in the text. Post-verdict scenarios, not gate evaluations.